Weight and local gravity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish mass from gravitational weight.

SUCCESS CRITERIA - I can...

1. Use mass in kg and force in N.
2. Calculate weight using the stated local g.
3. Explain how weight changes while mass stays the same.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 5 kg cart starts from rest under 35 N net force right for 4 s. Find acceleration and final velocity. State one assumption.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F_g = mg$; $g = F_g/m$.
F_g is weight. Field strength in N/kg is numerically equal to free-fall acceleration in m/s².

Worked reasoning: A 3 kg object in a field of 4 N/kg has weight 3×4=12 N. Moving it to a 2 N/kg field halves weight to 6 N but leaves mass 3 kg.

Your turn: Draw weight arrows for the two places and label the unchanged quantity.

Location	Mass (kg)	g (N/kg)	Weight (N)
A	3	4	12
B	3	2	6

Marzano check - circle: ① ② ③

03 Read and interpret the model

The same 2 kg object visits worlds with g=10 and g=5 N/kg. Compare its mass and weight.

World	g (N/kg)
A	10
B	5

Marzano check - circle: ① ② ③

04 Apply the physics

On World B, g = 5 N/kg. Complete the equipment table. Explain why doubling the packed mass doubles its weight.

Equipment mass (kg)	Weight (N)
3	____
6	____

Marzano check - circle: ① ② ③

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05 Explain your reasoning

A scale reports 0 N for an astronaut in an orbiting spacecraft. Does that prove Earth exerts no gravitational force? Explain.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 4 kg object has 32 N right and 8 N left. Draw the forces and find net force and acceleration.
Draw the forces on the chosen object and label their sources.

Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

A 4 kg object is where g=3 N/kg. State its mass and weight.

Add and label force arrows.

Draw and label the gravitational force.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two objects weigh the same on different worlds. Must their masses be the same? Build a numerical counterexample.

Case	Mass (kg)	g (N/kg)	Weight (N)
1	___	___	___
2	___	___	___

Marzano check - circle: 1 2 3